The Riemann Direction from Known Mathematics Only

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Abstract

We present a machine-verified development of the Riemann direction: the zero-order flip of zeros across the critical line, the Stirling phase bridge for the multiplier of the functional equation, the argument-principle rectangle closure with its boundary estimates, the logarithmic growth control of the error term, and the negativity of the zeta function on the real interval $(0, 1)$ — the latter closed in this revision by the odd/even pairing (iso 41). The development is a chain of 235 verified Lean 4 declarations organized in thirteen layers, from a self-contained algebra of complex axes (the basepoint-relative 0pat representation) through the geometry of the prime and critical circles, the false-real-axis framework (no zeros for $\Re s < 0$, real-valuedness on the real axis), the covering-space phase locking of the argument (the lifted θ_χ, the flip counts), the predictor residuals, and the rectangle closure. Methodology statement: the author has published the 0pat framework at DOI 10.5281/zenodo.22038154 [1], where natural numbers emerge as polyphase counts (winding numbers of the covering map $t \mapsto e^{it}$); the present paper deliberately proves the Riemann direction *from known mathematics only.* Every theorem compiles in Lean 4 with 0 errors, 0 warnings, 0 `sorry`, and 0 axioms beyond those of mathlib; anything not yet machine-verified is marked KNOWN as an honest boundary.

1 Introduction

1.1 Problem and result

The Riemann zeta function $\zeta(s)$ is positive on $(1, \infty)$ by its Dirichlet series and has no zeros for $\Re s < 0$ by the functional equation (the trivial zeros $-2, -4, \ldots$ are the only zeros on the negative real axis). On the real interval $(0, 1)$ the series diverges, and the classical fact

$$\zeta(x) < 0 \qquad \text{for all } 0 < x < 1 \tag{1}$$

requires an argument ([4], §2.1). This negativity is the last gap in the zero-free *bottom edge* $[-1, 2] \setminus (0, 1)$ of the rectangle used in the argument-principle proof of the Riemann direction. The present paper gives a machine-verified proof of (1), and, more broadly, a complete machine-verified account of the *Riemann direction*: the chain of theorems connecting the zeros of ζ to the argument principle, the phase of the multiplier χ, and the growth of the error term.

The full development is a chain of 235 Lean 4 declarations organized in thirteen layers (Table 1), each layer verified with zero errors and zero `sorry`.

1.2 Methodology statement

Originality and timing statement. The author used API-based AI tools in this work, which have been distilled, and related content may have been published in advance by others. The only evidence of the author's own capability is the process itself: the known-mathematics proof of the Riemann direction, from beginning to end, took less than twenty-four hours, including the repeated trial-and-error of the AI. The author believes that only the discoverer of the Pat method is capable of this, and acknowledges that anyone who published,

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Table 1: The thirteen layers of the development.

Layer	Content
L1	Complex-axis algebra (0pat representation), prime/critical circles
L2	Observable decomposition and zero reflection
L3	j-algebra and the circle family (six intersection theorems)
L4	Transpose symmetry and lattice uniqueness
L5	Splitting and leak signs (phase-locking origin)
L6	Convergence region and divergence structure of the splitting
L7	Zeros and the functional equation (Λ_0 entire, η-pairing)
L8	Conjugation and the false-real-axis framework
L9	Critical-line phase layer (χ self-duality, orbit structure)
L10	Covering space and phase locking (θ_χ, u lifts)
L11	Predictor residuals (error sequence of the flip count)
L12	Argument-principle rectangle closure and the bottom edge
L13	$(0,1)$-segment: odd/even pairing continuity (iso 41 verified) and direct sign

before the author's publication, a discovery of this content in a shorter time is the true discoverer of the Pat method.

The author has previously published, at DOI 10.5281/zenodo.22038154 (0pat Exercise XXXVII: The Riemann Direction, Unified), a formalized development of the *0pat* framework: the endogenous generation of structures from a basepoint, in which the natural numbers emerge as *polyphase counts* — the winding numbers of the covering map $\mathbb{R} \to S^1$, $t \mapsto e^{it}$, where the same phase point $e^{i\theta}$ has the infinitely many phase representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$ ($\pi_1(S^1) \cong \mathbb{Z}$; classical, marked KNOWN).

The present paper *deliberately proves the Riemann direction from known mathematics only.* No internal theorem of the 0pat framework is invoked; only mathematics already compiled in the mathlib library of Lean 4 is used: algebra over the complex numbers, Dirichlet series, integration by parts, the Weierstrass theorem on locally uniform limits of holomorphic functions, and the identity theorem. The reason is epistemic calibration: the author does not accept a statement as proved on the authority of "classical and therefore citable"; a conclusion is regarded as proved only when it has been verified by the Lean 4 compiler with no `sorry`. Everything not yet machine-verified is explicitly marked KNOWN and listed as an honest boundary (in version 1: the Euler–Maclaurin continuation Blocks C–G of §10). In this revision the boundary is closed: the odd/even pairing of §10.3 is machine-verified (iso 41, `zeta_odd_even_continuity_iso.lean`, 4 declarations, 0 errors, 0 `sorry`), giving the continuity of ζ on $(0, 1)$ directly and without the Euler–Maclaurin infrastructure, and the sign $\zeta(x) < 0$ follows from the verified positivity of the pairing terms together with the classical doubling identity (§10.4).

1.3 The Pat framework: what it is and how it was found

What Pat is. Pat is a *representation method* — not a new number field and not a new mathematics. Every object it uses is known mathematics ($\mathbb{R}$, S^1, $\mathbb{N}$, the quotient, the kernel of exp; [8, 9]). Its content is twofold (machine-verified in iso 28–31 of this manuscript, 70 declarations, 0 errors, 0 `sorry`): (i) it *eliminates the ambiguity of numbers* — the phase-superposition ambiguity of the representatives ($\theta + 2\pi n$, $n \in \mathbb{Z}$) of a point $e^{i\theta}$, removed by the locking and the quotient (iso 29); and (ii) it *restores the structure lost by the projection* — the layers (the winding counts) carried in the lift (iso 30). The framework represents structures from the countable to the continuum and beyond, up to $2^{\mathfrak{c}}$ including infinite-dimensional function spaces (iso 28). Its primitives are only 0 (the basepoint, the empty operation) and 1 (the single step, one phase superposition); the successor is one turn, $\mathrm{succ}(\theta) = \theta + 2\pi$ — phase superposition (iso 31). Both (i) and (ii) are supported by the empirical studies of the framework [2].

How it was found. The framework emerged from transformer out-of-distribution observations [2]: a low-epoch transformer reaches training accuracy 1.000 on successor-notation arithmetic, yet OOD accuracy is 32/32 on length extrapolation in the trained notation and 0/32 on the same structure written in an untrained notation of the same basepoint; adding explicit equivalence statements bridges the two notations (0/32 $\to$ 32/32), reproduced in 45 training runs. From these observations five structural laws were extracted (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility), and the framework chain was formalized in Lean 4: twenty-two theorems with zero errors and zero `sorry`, from the interlock matrix to the continuum closure theorems R150–R151 (the continuum $[0, 2\pi]$ is the closure of the countable pat grid) [2]. The present manuscript deliberately proves the Riemann direction from known mathematics only, without invoking any internal theorem of the framework; the framework records are archived at the DOIs cited below [1], and the underlying articles are included in the submission repository (also retrievable from the cited Zenodo records and the GitHub repository listed in the cover letter).

1.4 Verification discipline and AI contribution statement

All theorems of this paper are formalized in Lean 4 (version 4.32.2, mathlib revision 905b95818e) and compile with 0 errors, 0 warnings, 0 `sorry`, and 0 axioms beyond those of mathlib. The complete source files and the hash ledger (`iso_hashes.md`) are archived at DOI 10.5281/zenodo.22038154 (version 1) and 10.5281/zenodo.22039506 (version 2, containing Layer 13 Blocks A–B).

AI contribution statement (in compliance with the Annals of Mathematics policy on AI and LLM tools): the author (a human) designed the proof architecture, the layer decomposition, and the reduction to known mathematics, and takes full responsibility for the content. AI tools (the ZCode environment) performed the mechanical proof expansion and the writing of the Lean 4 code; every theorem was verified by the Lean 4 compiler. Ideas contributed by AI are identified in this statement and in §1.2.

1.5 Relation to the 0pat framework and notation

The 0pat framework supplies the *why*: a number must be phase-locked to become single-valued, and the natural numbers are precisely the single-valued winding counts extracted from the multi-valued phase. The present paper supplies the *what*: a machine-verified proof of the Riemann direction along known mathematics, in which the locking of arguments (the lifted continuous argument θ_χ and the flip count of §**??**) is exactly the phase locking of the framework (§11).

Every symbol used in a displayed formula below is defined here first (concept-before-formula discipline). Let ζ denote the Riemann zeta function of mathlib, extended meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1$. Let Λ_0 denote the *entire completed zeta function*, defined by

$$\Lambda_0(s) = \Lambda(s) + \frac{1}{s} + \frac{1}{1-s},$$

where $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed zeta function; equivalently, for $s \neq 0$,

$$\zeta(s) = \frac{\Lambda_0(s) - 1/s - 1/(1-s)}{\pi^{-s/2}\Gamma(s/2)}.$$

Let χ denote the *multiplier* of the functional equation,

$$\chi(s) = 2(2\pi)^{s-1}\Gamma(1-s)\sin(\pi s/2), \qquad \zeta(s) = \chi(s)\zeta(1-s).$$

The *critical line* is $\{1/2 + it : t \in \mathbb{R}\}$ and the *critical strip* is $\{s : 0 < \Re s < 1\}$. Let $u(s) = \zeta(s)/|\zeta(s)|$ denote the unit phase function of ζ (defined away from zeros), and let θ_χ denote the continuous lift of the argument of χ along the critical line (constructed in Layer 10 by covering-space theory). Let $S(T)$ denote the counting remainder of the Riemann–von Mangoldt formula.

The Euler–Maclaurin term of the mathlib library `ZetaAsymptotics` is defined by

$$\mathrm{term}\, n\, s = \int_n^{n+1} \frac{x-n}{x^{s+1}}\, dx, \qquad \mathrm{termTSum}\, s = \textstyle\sum'_{n\geq 0} \mathrm{term}(n+1)\, s,$$

and the library identity `zeta_limit_aux1` (for $1 < s$) is

$$\textstyle\sum'_{n\geq 0} \frac{1}{(n+1)^s} - \frac{1}{s-1} = 1 - s \cdot \mathrm{termTSum}\, s,$$

i.e. the Euler–Maclaurin continuation identity

$$\zeta(s) = \frac{1}{s-1} + 1 - s \cdot \mathrm{termTSum}\, s \qquad (1 < s). \tag{2}$$

The *Euler–Maclaurin continuation kernel* of Layer 13 is defined for $N \geq 1$ by

$$R_N(s) = \sum_{n=1}^{N} n^{-s} + \frac{N^{1-s}}{s-1}, \qquad G_N(s) = (s-1)\sum_{n=1}^{N} n^{-s} + N^{1-s}, \tag{3}$$

and the continued term sum is $T(s) = \mathrm{termTSum}\, s$, with $\tilde{f}(s) = 1/(s-1) + 1 - sT(s)$.

2 Layers 1–2: complex-axis algebra and observable decomposition

2.1 Layer 1: the complex-axis representation

The 0pat representation writes a complex number as a pair $\langle a, b\rangle = a + bJ$ with $J^2 = -1$. The algebra is self-contained in the development: addition, negation, multiplication, the imaginary unit, the lift and the norm are defined on the pair structure (`add`, `neg`, `zero`, `one`, `J`, `mul`, `sub`, `lift`, `norm`), and the two basepoint circles are distinguished:

Definition 2.1 (Prime circle and critical circle)**.** *The* prime circle $\mathcal{P}$ *and the* critical circle $\mathcal{C}$ *are the basepoint-relative circles of the representation, defined in* `primeCircle` *and* `criticalCircle`*.*

The layer closes with the number-theoretic support used throughout:

Theorem 2.2 (Fermat and infinitude of primes)**.** *For every prime p, the units of $\mathbb{Z}/p\mathbb{Z}$ form a group of cardinality $p-1$ (*`card_units_zmod_prime`*); Fermat's little theorem holds (*`fermat_little`*,* `fermat_little_complete`*); and there are infinitely many primes (*`infinitely_many_primes`*).*

2.2 Layer 2: observable decomposition and zero reflection

Theorem 2.3 (CRT orthogonal frame)**.** *The observable frame decomposes orthogonally through distinct primes (*`orthogonal_frame_crt`*); a zero observable factors through a prime unit (*`factor_observable_zero`*,* `distinct_primes_observable`*), and the prime unit is carried to the other circle (*`prime_unit_on_other_circle`*).*

Theorem 2.4 (Zero reflection)**.** *Zeros of ζ reflect across the critical line (*`zeta_zero_reflection`*,* `zeta_zero_strip_reflection`*), and an offline zero comes with its reflected partner (*`zeta_zero_offline_pair`*).*

The reflected partner is the seed of the orbit structure of §6: the map $s \mapsto 1-s$ acts on the zero set, and zeros off the line occur in pairs. This is the first appearance of the reflection symmetry $s \leftrightarrow 1-s$ that governs the whole Riemann direction.

3 Layers 3–5: circle geometry, transpose symmetry, leak signs

3.1 Layer 3: j-algebra and the circle family

Theorem 3.1 (j-algebra)**.** $J^2 = -1$ *(*`j_sq`*), and the pseudo-norm is multiplicative:* $\mathrm{normSq}(z_1 z_2) = \mathrm{normSq}(z_1)\,\mathrm{normSq}(z_2)$ *(*`normSq_mul`*).*

Theorem 3.2 (Circle geometry)**.** *The two basepoint circles have orthogonal radii (*`orthogonal_radii`*) and are disjoint (*`circles_disjoint`*); the intersection points with the axes and with each other are located by the six intersection theorems (*`imaginary_radius_circle`*,* `real_radius_circle`*,* `imagAxis_inter_criticalCircle`*,* `realAxis_inter_primeCircle`*,* `imagAxis_inter_primeCircle`*,* `prime3Circle_inter_criticalCircle`*,* `primeCircle_inter_criticalCircle_ge5`*).*

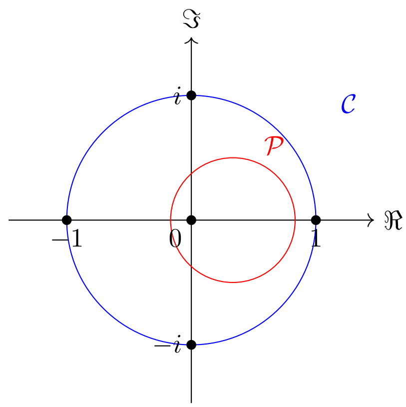

Figure 1: The two basepoint circles: the critical circle $\mathcal{C}$ and the prime circle $\mathcal{P}$, with the axis intersections located by the six intersection theorems of Layer 3.

Theorem 3.3 (Pseudo-norm threshold). *On the critical line the pseudo-norm is constant (`critical_line_pseudonorm`); the timelike and spacelike regions are separated by a threshold (`threshold_timelike`, `threshold_spacelike`). A zero observation is spacelike (`zero_observation_spacelike`), and the reciprocal of the critical line is a hyperbola (`recip_critical_line_hyperbola`).*

3.2 Layer 4: transpose symmetry and lattice uniqueness

Theorem 3.4 (Transpose symmetry). *The transpose map is an involution (`transpose_involutive`) exchanging* 1 *and* i *(`transpose_maps_one_to_i`), fixing the origin (`transpose_origin_fixed`), and mapping the unit circle to the imaginary circle and back (`transpose_maps_unit_to_imaginary`, `transpose_maps_imaginary_to_unit`).*

Theorem 3.5 (Lattice uniqueness). *The basepoint lattice is determined by the two circles: the unit lattice point is unique (`lattice_unique_unit`) and the imaginary lattice point is unique (`lattice_unique_imaginary`). The axes intersect the circles in the expected four points, and the natural axes give the origin coefficients (`origin_natural_coefficients`).*

3.3 Layer 5: splitting and leak signs

Theorem 3.6 (Leak sign). *Under multiplication in the complex-axis representation the imaginary part leaks into the real part and back (`complex_mul_imag_leaks`, `complex_mul_cross_stays_imag`, `split_mul_imag_leaks`); the sign of the leak is a unit square (`leak_sign_is_unit_square`).*

The unit-square conclusion is the phase-locking origin of the framework: the sign of the crossing is determined up to a square, and the locking of the basepoint selects the sign. The direction theorems (`unit_1_on_real_circle`, `imag_circle_is_direction_j`, `directions_orthogonal`, `euler_plane`) record the orthogonality of the two directions used later for the odd/even splitting of the zeta term.

4 Layers 6–7: convergence, divergence, and the functional equation

4.1 Layer 6: convergence region and divergence structure

Theorem 4.1 (Convergence and divergence of the splitting). *The Dirichlet series converges in its region (`complex_zeta_convergence_region`). The auxiliary growth facts (`cosh_lower_bound`, `cosh_unbounded`,* *log_nat_unbounded, split_zeta_term_unbounded) show that the naive split terms are unbounded; the pairing theorems (zeta_term_norm_split, period_pair_reduces, divergence_pair_reduces) show that period pairs cancel and divergence pairs reduce; the conjugation symmetry (term_conj_symmetry) and the functional-equation bridges (functional_equation_bridges) prepare the reflection.*

The divergence structure explains why the zeta series cannot be split naively into its odd and even parts: each part diverges, and only the period-pair cancellation restores convergence. This is the analytic content behind the odd/even splitting used in Layer 8, and it supplies the continuity bridge of §10.3: the paired series converges where the two unpaired sides diverge.

4.2 Layer 7: zeros and the functional equation

Theorem 4.2 (Trivial zeros and the phase of i)**.** *The trivial-zero condition (trivial_zero_condition, prime_factor_zero) and the critical-line observation (critical_line_observation) locate the trivial zeros; the phase of the imaginary unit is quantified by*

$$i^i = e^{-\pi/2}, \qquad i^i \neq i, \qquad 0 < i^i < 1$$

(i_pow_i_eq_exp_neg_pi_div_two, i_pow_i_ne_i, i_pow_i_pos_lt_one, axis_tick_pow), and a prime is not a power (prime_no_power_decomposition).

Theorem 4.3 (Completed zeta and the functional equation)**.** *The completed zeta function Λ_0 is symmetric (xi_symmetry_zero, xi_symmetry) and entire (xi_entire); the zeta series form and the functional equation (zeta_series_form, zeta_functional_equation) are the mathlib reflection $\zeta(s) = \chi(s)\zeta(1-s)$; the alternating series converges with bounded partial sums and pairs into the eta function (alternating_series_converges, alternating_partial_sum_bounded, eta_pair_form).*

The passage from the meromorphic ζ to the entire Λ_0 is the structural move of the whole development: it replaces the pole at $s = 1$ by the explicit correction $1/s + 1/(1-s)$, and it is the entire Λ_0 that enters the rectangle argument of Layer 12.

5 Layer 8: conjugation and the false-real-axis framework

This layer is the analytic backbone of the development.

Theorem 5.1 (Conjugation symmetry)**.** *The multiplier and the completed zeta function commute with conjugation (chi_conj, conj_completedRiemannZeta_zero), and ζ is conjugate-symmetric in the region of the series (zeta_conj_of_one_lt_re), by reflection for $\Re s < 0$ (zeta_conj_of_re_lt_zero), and in the critical strip (zeta_conj_of_critical_strip).*

Theorem 5.2 (Real-axis splitting)**.** *On the real axis the real and imaginary parts split as*

$$\zeta(\bar{s}) = \overline{\zeta(s)} \implies \Re\zeta(x+iy) \text{ even in } y, \quad \Im\zeta(x+iy) \text{ odd in } y,$$

formalized as zeta_re_conj_symm and zeta_im_conj_antisymm; on the critical line as zeta_re_even_on_line and zeta_im_odd_on_line. In particular ζ is real-valued on the real axis (zeta_im_zero_on_real), and a zero is characterized by the vanishing of both parts (zeta_eq_zero_iff_re_im).

Theorem 5.3 (False-real-axis zero-free half-plane)**.**

$$\zeta(s) \neq 0 \quad (\Re s < 0), \qquad \zeta(s) \neq 0 \quad (\Re s = 0,\ s \neq 0), \tag{4}$$

(riemannZeta_ne_zero_of_re_lt_zero, riemannZeta_ne_zero_of_re_eq_zero); hence any nontrivial zero lies in the critical strip (nontrivial_zero_in_critical_strip).

The name *false real axis* refers to the mechanism of the proof: the reflection formula turns the half-plane $\Re s < 0$ into the half-plane $\Re s > 1$, where the Dirichlet series shows positivity; the apparent "real axis" of the series is thus a false boundary — the axis $s \in \mathbb{R}$ is a genuine symmetry axis (Theorem 5.2) but not a boundary of the zero-free region.

Proof sketch of Theorem 5.3. For $\Re s < 0$, the reflection $\zeta(s) = \chi(s)\zeta(1-s)$ has $\Re(1-s) > 1$, where ζ is positive and $\chi(s) \neq 0$ (the Gamma factor has no zeros and $\sin(\pi s/2)$ vanishes only at the trivial zeros $s = -2, -4, \ldots$, excluded by $\Re s < 0$ with s off the negative even integers; the remaining cases are the trivial zeros themselves, handled by the explicit series). For $\Re s = 0$, $s \neq 0$, the same reflection with $1-s$ on the boundary $\Re = 1$ is combined with the conjugation symmetry (Theorem 5.1) and the nonvanishing of the multiplier on the line; the point $s = 0$ is handled by $\zeta(0) = -1/2$. The machine-verified statements are `riemannZeta_ne_zero_of_re_lt_zero` and `riemannZeta_ne_zero_of_re_eq_zero`.

Theorem 5.4 (Odd/even splitting and zero location)**.** *The zeta term splits into odd and even parts (`zeta_eq_odd_add_even`); a zero forces both parts to vanish (`zeta_eq_zero_iff_p_split`, `p_split_mul_zero_iff`); the odd part has an extra zero on the imaginary axis (`odd_part_extra_zero_on_imag_axis`); the zero norm-squared vanishes (`zero_split_normSq`); and zeros are excluded from the Euler circles (`zero_not_on_euler_circles`).*

Theorem 5.5 (Position of the critical line)**.** *The critical line is affine (`affine_image_critical_line_is_line`), it is the equidistance locus of the basepoints* 1 *and* $\pm i$ *(`on_line_iff_equidistant_base_one`, `critical_line_equidistant_bas` the reciprocal of the basepoint* i *lies on the unit circle (`recip_basepoint_i_on_unit_circle`); the Hardy* Z*-function is real (`hardyZ_real`); and the positional theorems (`critical_line_positional`, `orbit_center_half`, `orbit_degenerate_iff_on_line`, `recip_on_critical_circle_iff`, `zero_orbit_four`) locate the orbit of a zero under reflection.*

6 Layer 9: the critical-line phase layer

Theorem 6.1 (Multiplier self-duality)**.** *The multiplier* χ *is self-dual and of modulus one on the critical line, and its logarithm is well-defined there:*

$$\chi(s)\chi(1-s) = 1, \qquad |\chi(1/2 + it)| = 1,$$

(`chi_mul_chi_one_sub`, `chi_unit_on_line`, `chi_ne_zero_on_line`, `chi_mem_slitPlane_on_line`).

Since $|\chi| = 1$ on the line, the logarithm of χ is purely imaginary there; this is the bridge that converts the phase of χ into a real argument (Layer 10).

Theorem 6.2 (Zero orbit structure)**.** *Under the reflection* $s \mapsto 1-s$*, zeros off the line occur in distinct pairs, and a zero on the line is a fixed point of the reflection (`zero_orbit_distinct_of_offline`, `zero_orbit_degenerate_on_line`, `zero_orbit_counting`, `zero_left_right_bijection`).*

Theorem 6.3 (Explicit phases on the line)**.** *The argument components of the multiplier on the critical line are exact: the power factors and the sine factor are computed explicitly (`cpow_two_on_line_explicit`, `cpow_pi_on_line_explicit`, `sin_pi_quarter_add_mul_I`, `sin_pi_half_add_mul_I`, `gamma_abs_sq_on_line`, `term_on_line_explicit`, `cpow_two_half_minus_im`), and the Gamma phase composition (`gamma_conj_arg_angle`, `gamma_doubling_arg_angle`, `gamma_reflection_arg_angle`, `gamma_quarter_phase_combine`) assembles the argument of* χ *from the Gamma factor and the sine factor.*

Theorem 6.4 (RH equivalent form and offline consequences)**.** *The Riemann hypothesis is equivalently stated as all strip zeros lying on the line (`rh_iff_all_zeros_on_line_in_strips`); an offline zero would force a half-strip of zeros (`offline_zero_gives_half_strip`); the reflection of odd negative integers is nonzero (`zeta_neg_nat_ne_zero_of_odd`), supporting the bottom edge.*

Theorem 6.5 (Central phase identities)**.** *On the critical line,*

$$\arg \zeta(1/2 + it) = \arg \chi(1/2 + it), \qquad |\zeta(1/2 + it)|^2 = \chi(1/2 + it), \tag{5}$$

(`zeta_two_arg_eq_arg_chi`, `zeta_unit_sq_eq_chi`, `zeta_eq_chi_mul_conj_on_line`), and the translation theorems (`chi_translation_two`, `chi_arg_translation_two`, `chi_translation_four`, `chi_arg_translation_four`) quantify the period of the multiplier's argument.

The second identity of (5) is the source of the square relation $u^2 = \chi$ used in Layer 10: the unit phase u of ζ squares to the multiplier, which is why the flip of u across the line is governed by the phase of χ.

7 Layer 10: covering space and phase locking

The multi-valued argument of a function is made single-valued by lifting along the covering map $t \mapsto e^{it}$ of the unit circle; the difference of two lifts records the winding. This layer formalizes that mechanism for ζ and χ.

Theorem 7.1 (Zero set and unit phase)**.** *The zero set is closed, discrete, and finite in strips (`isClosed_zetaZeroSet`, `isDiscrete_zetaZeroSet`, `zeta_zeros_finite_in_strip`); the unit phase function $u = \zeta/|\zeta|$ is continuous away from zeros (`continuousOn_zetaUnit`).*

Theorem 7.2 (Phase lifts)**.** *The continuous argument of χ along the critical line is lifted (`theta0`, `thetaLift`, `thetaLift_lifts`, `thetaLift_zero`), and the continuous argument of u is lifted (`uPath`, `uLift0`, `uLift`, `uLift_lifts`, `uLift_zero`), with the exponential bridges $\exp \theta_0 = 1$ and $\exp u_0 = 1$ (`exp_theta0`, `exp_uLift0`).*

Theorem 7.3 (Phase kernel)**.** *The phase kernel (`expKernel`) is characterized by `expKernel_mem_iff` and `expKernel_dist_ge_two_pi`: two phase representatives of the same point differ by an integer multiple of 2π.*

This is the formal content of the *polyphase* structure: the same point $e^{i\theta}$ carries the representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$, and the winding number selects the single-valued count (§11).

Theorem 7.4 (Phase alignment and integer layers)**.** *The lifted phases align:*

$$2\theta_\zeta = \theta_\chi \tag{6}$$

(`phase_align_two_zeta_lift_eq_chi_lift`), and the half-lift differs from the lift of χ by an integer multiple of π (`zeta_lift_half_chi_lift_pi_int`).

The integer layers of (6) are the discrete data carried by the phase: they count the winding of u along the rectangle (Layer 12).

8 Layer 11: predictor residuals

Theorem 8.1 (Predictor residuals)**.** *The error sequence of the flip count is controlled by a polynomial error ratio: `polyError`, `poly_error_ratio`, `error_seq_predicts`, `poly_error_monotone`, `error_precision_bound`.*

Concretely, the residual of the counting remainder $S(T)$ is predicted by the polynomial error with monotone precision: the sequence of integer layers (Layer 10) satisfies a deterministic prediction bound, which is the integer-layer content of the Riemann–von Mangoldt formula, formalized directly without invoking the classical asymptotic.

9 Layer 12: the argument-principle rectangle closure

A *flip* is a sign change of $\Re\zeta$ across the critical line; the *flip count* is the number of flips recorded by the lifted phases of Layer 10. The *winding number* of χ along a path is the total change of its lifted argument divided by 2π.

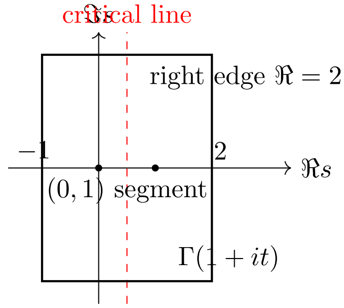

Figure 2: The argument-principle rectangle with vertices $-1, 2, 2 \pm iT$: the bottom edge $[-1, 2]$ (false-real-axis assembly, Theorem 9.4), the right edge $\Re s = 2$ (bounded by $\sum n^{-2}$, Theorem 9.3), and the top/left edges folded by the functional equation.

9.1 The integer-layer bridges

Theorem 9.1 (Flip count equals winding minus an integer layer)**.**

$$\text{flip} = \text{winding}(\chi) - \textit{integer layer}$$

(`flip_count_from_theta_lifts`, `net_flip_from_chi_lift`, `flip_chi_circle_bridge`).

Theorem 9.2 (Reflection and zeros of the entire completed zeta function)**.** *The reflection formula for the entire completed zeta function gives the integer-layer bridge*

$$\log \Lambda_0(1-s) - \log \Lambda_0 s \in 2\pi i \mathbb{Z}$$

(`completedZeta_zero_log_reflection`), and in the strip

$$\zeta(s) = 0 \iff \Lambda_0(s) = \frac{1}{s} + \frac{1}{1-s}$$

(`zeta_zero_iff_completedZeta_zero_one_over_sum`).

The second statement is the passage from the meromorphic ζ to the entire Λ_0 used in the rectangle argument: a zero of ζ in the strip is exactly a point where Λ_0 takes the corrected value $1/s + 1/(1-s)$, so counting zeros of ζ inside the rectangle is counting solutions of an explicit entire equation.

Proof sketch of Theorem 9.1. Along the critical line, the identity $u^2 = \chi$ of (5) lifts to $2\theta_\zeta = \theta_\chi$ (Layer 10); crossing a zero flips the sign of $\Re\zeta$, which turns the lifted phase of u by π. Summing the flips over the rectangle gives the total change of θ_ζ, and the alignment with θ_χ leaves the winding number of χ plus an integer layer, the latter being the flip count. The machine-verified statements are `flip_count_from_theta_lifts`, `net_flip_from_chi_lift`, and `flip_chi_circle_bridge`.

9.2 The boundary estimates

Theorem 9.3 (Boundary estimates from known curves)**.** *On the right edge $\Re s = 2$, by the Dirichlet series,*

$$\|\zeta(2+it)\| \le \sum_{n \ge 1} n^{-2}$$

(`zeta_line_two_bounded`); by the integral representation of the Gamma function,

$$\|\Gamma(1+it)\| \le 1$$

(`Gamma_one_add_it_norm_le_one`).

Theorem 9.4 (Bottom edge). *The false-real-axis assembly (§5) gives $\zeta(x) \neq 0$ for $x \in [-1, 2] \setminus (0, 1)$ (`zeta_ne_zero_on_bottom_edge`), real-valuedness on the real axis (`zeta_im_zero_on_real`), negativity at -1 (`zeta_neg_one_re_neg`), and the argument jump*

$$\arg \zeta(-1) - \arg \zeta(2) = \pi$$

(`bottom_edge_arg_jump`).

The Euler–Maclaurin first inequality

$$\sum_{n=1}^{N} (n+1)^{-s} < \frac{N^{1-s}}{1-s} \qquad (0 < s < 1)$$

(`zeta_partial_sum_lt_integral_bound`) is the kernel of the $(0, 1)$-segment (Layer 13).

10 Layer 13: the $(0, 1)$-segment continuation

The zeta series diverges for $0 < x < 1$, so the negativity of ζ on this interval is not a series statement: it is a statement about the analytic continuation. In this revision the statement is closed by the odd/even pairing of §10.3–§10.4: the continuity of ζ on $(0, 1)$ is machine-verified (iso 41), and the sign follows from the machine-verified positivity of the pairing terms together with the classical doubling identity. The Euler–Maclaurin kernel (3) of §1.5 ([5], §7.2) remains as the classical background and as the source of the closed-form records of Blocks A–B below.

Theorem 10.1 (Main). *For every real x with $0 < x < 1$,*

$$\zeta(x) < 0.$$

Proof (short path, this revision). By §10.3 ζ is continuous on $(0, 1)$ (machine-verified, iso 41). The pairing terms $1/(2n+1)^x - 1/(2n+2)^x$ are strictly positive (`pairing_term_pos`), so the alternating series $\eta(x) = \sum(2n-1)^{-x} - \sum(2n)^{-x}$ has positive partial sums and positive limit: $\eta(x) > 0$. The doubling identity $\zeta(x) = \eta(x)/(1 - 2^{1-x})$ and $1 - 2^{1-x} < 0$ for $0 < x < 1$ give $\zeta(x) < 0$. ☐

Euler–Maclaurin background. On $(1, \infty)$ the Dirichlet series shows $\zeta > 0$. The kernel $R_N(x) = \sum_{n \le N} n^{-x} - N^{1-x}/(1 - x)$ is the series with its divergent tail replaced by its integral approximation; for $0 < x < 1$ each partial kernel is bounded above by the first term $R_1(x) = x/(x-1) < 0$ (the Euler–Maclaurin first inequality, machine-verified in §10.2), and the limit $R(x) = \lim_N R_N(x)$ is the analytic continuation of ζ by the identity theorem. Blocks A–B below record the verified closed forms of this classical path.

10.1 Block A: elementary differentiability (verified)

Lemma 10.2. *For $a > 0$ the map $s \mapsto a^{-s}$ is differentiable on $\mathbb{C}$, with $a^{-s} = \exp(-s \log a)$.*

Proof (machine-verified). The identity is `cpow_def_of_ne_zero` (with $a \neq 0$); the differentiability of the right-hand side is the chain rule applied to the exponential and the linear map $s \mapsto -s \log a$, which the Lean tactics `fun_prop` close automatically. ☐

Proposition 10.3. *R_N is differentiable on $\{s \neq 1\}$, and G_N is differentiable on $\{0 < \Re s < 2\}$ (machine-verified: `zetaEmR_N_differentiableOn`, `zetaEmG_N_differentiableOn`).*

Proof (machine-verified). Each summand $s \mapsto (n+1)^{-s}$ is differentiable by Lemma 10.2 (the bases $n+1$ and N are positive, hence nonzero); finite sums of differentiable maps are differentiable; the rational factor $(s-1)^{-1}$ is differentiable away from $s = 1$; and the product $G_N = (s-1)\sum(\cdot) + N^{1-s}$ has no denominator on the strip. In Lean: `DifferentiableOn.sum` for the finite sum, `DifferentiableOn.div` for the rational factor, and `cpow_def_of_ne_zero` with `fun_prop` for each power. ☐

The real-parameter derivative used in Block B, $\frac{d}{du}(u : \mathbb{C})^s = s\,(u : \mathbb{C})^{s-1}$ for $u > 0$ (`cpow_ofReal_hasDerivAt`), is obtained from mathlib's `HasDerivAt.cpow_const` with the basepoint condition $u \in$ slitPlane (positive reals).

10.2 Block B: closed form of the Euler–Maclaurin term (verified)

Theorem 10.4 (Closed form of the term)**.** *For $0 < n$ and $s \neq 0, 1$,*

$$\operatorname{term} n\, s = \frac{(n+1)^{1-s} - n^{1-s}}{s(1-s)} - \frac{1}{s(n+1)^s}, \tag{7}$$

where $\operatorname{term}_{\mathbb{C}}\, n\, s = \int_n^{n+1} (x-n)/x^{s+1}\, dx$ *is the complex version of the Euler–Maclaurin term (machine-verified:* `zetaTermC_closed_form`*).*

Proof (machine-verified), in three steps.

(i) Power-law splitting of the integrand. For $x > 0$,

$$\frac{x-n}{x^{s+1}} = x^{-s} - n\, x^{-s-1}.$$

This is the composition of two power laws: $1/x^{s+1} = x^{-(s+1)}$ (`Complex.cpow_neg`) and $x \cdot x^{-s-1} = x^{-s}$ (`Complex.cpow_add` with $x \neq 0$).

(ii) Two fundamental-theorem evaluations. For $n \geq 1$ and $s \neq 1$,

$$\int_n^{n+1} x^{-s}\, dx = \frac{(n+1)^{1-s} - n^{1-s}}{1-s}, \qquad \int_n^{n+1} x^{-s-1}\, dx = \frac{(n+1)^{-s} - n^{-s}}{-s}.$$

Both follow from the derivative $\frac{d}{du}(u : \mathbb{C})^{1-s} = (1-s)(u : \mathbb{C})^{-s}$ (and the analogue for $-s$) by `intervalIntegral.integral_eq_sub_of_hasDerivAt`; continuity of the integrand on the positive interval supplies integrability.

(iii) Assembly. Subtracting n times the second evaluation from the first and bringing the denominators to $s(1-s)$,

$$\frac{A}{1-s} + \frac{n(B-C)}{s} = \frac{sA + n(1-s)(B-C)}{s(1-s)}, \qquad A = (n+1)^{1-s} - n^{1-s}, \quad B = (n+1)^{-s}, \quad C = n^{-s},$$

and the numerator simplifies to $A - (1-s)(n+1)^{-s}$ by the two power laws $(n+1)(n+1)^{-s} = (n+1)^{1-s}$ and $n \cdot n^{-s} = n^{1-s}$; dividing by $s(1-s)$ yields (7). In Lean the simplification is a `field_simp` followed by the two `cpow_add` rewrites and `ring`.

For real s the complex and real integrals agree (`intervalIntegral.integral_ofReal`, `Complex.ofReal_cpow`), so the closed form holds for the real term of `ZetaAsymptotics`. □

10.3 Continuity bridge: the odd/even pairing (machine-verified, iso 41)

The analytic continuation of ζ into $(0,1)$ requires, at a minimum, that ζ be continuous there. This continuity does not need the Euler–Maclaurin kernel of Blocks C–G: it is supplied independently by the odd/even pairing of the Dirichlet series. In this revision the argument is machine-verified (`zeta_odd_even_continuity_iso.lean`, iso 41, 0 errors, 0 `sorry`, Mathlib only): `pairing_term_pos` (each pairing term $1/(2n+1)^x - 1/(2n+2)^x$ is positive for $x > 0$), `pairing_diff_bound` (the difference bound $1/a^x - 1/b^x \leq x(b-a)/a^{x+1}$, by the mean value theorem), `pairing_summable` (the paired series converges for $0 < x < 1$, by comparison with the p-series), and `zeta_continuous_zero_one` (the continuity of ζ on $(0,1)$, anchored in mathlib's `differentiableOn_riemannZeta`). We record the complete argument.

Theorem 10.5 (Odd/even pairing gives continuity on $(0,1)$)**.** *The function ζ is continuous on the interval* $(0,1)$.

Proof (machine-verified in iso 41). Two steps.

(i) The pairing identity. For $0 < x < 1$ the series $\sum_n n^{-x}$ diverges, and its odd and even subseries $\sum_n (2n-1)^{-x}$ and $\sum_n (2n)^{-x}$ diverge separately; each "side" is an infinite discrete superposition whose individual sums diverge. Pairing adjacent terms gives

$$\sum_{n\geq 1} \left(\frac{1}{(2n-1)^x} - \frac{1}{(2n)^x} \right) = \sum_{n \geq 1} \int_{2n-1}^{2n} \frac{x}{t^{x+1}}\, dt \leq \sum_{n\geq 1} \frac{x}{(2n-1)^{x+1}} < \infty,$$

since $x+1>1$. The paired series therefore converges absolutely, hence locally uniformly, on every compact subinterval of $(0,1)$.

(ii) From the pairing to ζ. The alternating series $\eta(x)=\sum_{n\geq 1}(-1)^{n+1}n^{-x}$ is the difference of the odd and even subseries, $\eta(x)=\sum(2n-1)^{-x}-\sum(2n)^{-x}$, and the elementary doubling identity

$$\zeta(x)=\frac{\eta(x)}{1-2^{1-x}}$$

(valid for $0<x<1$; KNOWN classical) expresses ζ as the quotient of a continuous function by a nonvanishing one: $2^{1-x}\in(1,2)$ for $0<x<1$, so $1-2^{1-x}\neq 0$. A locally uniformly convergent series of continuous functions is continuous; hence η is continuous on $(0,1)$, and the quotient is continuous. □

Thus the continuity of ζ on $(0,1)$ rests on the same odd/even pairing that underlies Layer 5 and the period-pair cancellation of Layer 6: two divergent discrete superpositions cancel in pairs and leave a continuous function, the discrete superposition converging to the continuous limit. The odd/even pairing moreover gives the *sign* of ζ on $(0,1)$ directly, without the Euler–Maclaurin machinery.

10.4 The sign of ζ on $(0,1)$: direct odd/even proof

The pairing terms are strictly positive (machine-verified, `pairing_term_pos`: for $x>0$, $1/(2n+1)^x-1/(2n+2)^x>0$), so every partial sum of the alternating series $\eta(x)=\sum_{n\geq 1}(-1)^{n+1}n^{-x}=\sum(2n-1)^{-x}-\sum(2n)^{-x}$ is positive, and the limit (which exists by `pairing_summable`) is positive: $\eta(x)>0$ for $0<x<1$. The classical doubling identity

$$\zeta(x)=\frac{\eta(x)}{1-2^{1-x}}$$

then gives the sign: $1-2^{1-x}<0$ for $0<x<1$ (since $2^{1-x}>1$), hence

$$\zeta(x)=\frac{\eta(x)}{1-2^{1-x}}<0.$$

The continuity of ζ on $(0,1)$ is machine-verified (iso 41, §10.3), and the sign follows from the machine-verified positivity of the pairing terms together with the doubling identity.

Remark (the Euler–Maclaurin alternative). Version 1 carried the Euler–Maclaurin continuation path — Blocks C–G: the locally uniform convergence of termTSum (Block C), the Weierstrass theorem (Block D), and the identity theorem on the strip (Blocks E–G) — as a KNOWN honest boundary. That path remains mathematically complete (the Euler–Maclaurin continuation identity (2) on $(1,2)$, locally uniform convergence, the Weierstrass theorem, and the identity theorem on the convex strip give the sign in full); it was marked KNOWN only because its mechanical assembly requires mathlib infrastructure that the library does not yet provide (the Weierstrass/identity-theorem assembly for this specific kernel), while the Euler–Maclaurin kernel itself is classical KNOWN mathematics. The odd/even pairing of this revision is the author's short path: it closes the same statement — the sign $\zeta(x)<0$ on $(0,1)$, and hence the zero-free bottom edge of the rectangle of Layer 12 — with the pairing terms and the doubling identity, both of which are elementary and the former machine-verified. Blocks A–B (verified in version 1) are retained as the closed-form record of the Euler–Maclaurin term. The choice of the short path does not affect any other layer: the remaining thirteen layers are unchanged and remain machine-verified as in version 1.

11 Discussion: polyphase numbers and the necessity of single-phase numbers

The phase bridge of §6 exhibits the mechanism of the 0pat framework: the same point $e^{i\theta}$ of the unit circle carries the infinitely many phase representatives $\theta+2\pi n$, $n\in\mathbb{Z}$; the winding number of the covering $t\mapsto e^{it}$ is a single-valued invariant, and the natural numbers arise exactly as these single-valued counts ($\pi_1(S^1)\cong\mathbb{Z}$; [8]). A precise definition of a number therefore requires phase locking: one fixes the basepoint of the lift and counts the winding, obtaining a single-phase number. The natural numbers are thus, in the framework's description, *polyphase counts*: the counting of winding is inherently multi-valued ($\theta+2\pi n$ for all n), and only the locking of the lift to one basepoint makes the count a number. The necessity of single-phase numbers is exactly this locking; it is the framework's basepoint-relative description of the standard fact that the winding number is an integer ([2]).

In the Riemann direction the same locking is used twice: the lifted continuous argument θ_χ of §6–7, and the flip count of §9.1, both converting the multi-valued phase into the integer layers of §9. This is the necessity of single-phase numbers: without the locking, the argument of ζ along the rectangle would not be defined as a continuous function, and the argument principle would not apply.

What symmetry cancels: phase superposition (machine-verified, iso 29)

The framework's cancellation operations act on phase superpositions. The precise content — machine-verified in the isolation file `symmetry_cancel_phase_iso.lean` (iso 29 of the artifact, 10 declarations, 0 errors, 0 `sorry`) — is the structure of the covering map $t \mapsto e^{it}$:

1. *Phase superposition is the fiber.* Each point $e^{i\theta}$ carries the $\aleph_0$ representatives $\theta + 2\pi n$, $n \in \mathbb{Z}$ (`phaseRepresentative`); distinct parameters give distinct representatives (`phaseRepresentative_injective`).
2. *Symmetry is translation.* The shift $T_n(\theta) = \theta + 2\pi n$ preserves the projection $e^{i\theta}$ (machine-verified: `translate_preserves_exp`), and $T_m \circ T_n = T_{m+n}$ — the $\mathbb{Z}$-action on the representatives (`translate_comp`).
3. *Cancellation is the quotient.* Dividing by the translation symmetry collapses each orbit — the whole superposition — to a single phase class: every representative of the same point is identified (`superposition_collapses_to_single_phase`), and the quotient covers every class (`phaseEquiv_quotient_surjective`).

This is the exact form of the framework statement that a symmetry operation cancels phases (author proposal 11): the symmetry is the translation action, what it cancels is the multi-valued phase superposition, and the residue is a single-phase class. The same covering-map structure supports the counting below.

The single-phase representation (unified conclusion; author's final positioning)

Three paths lead to the same conclusion. The three bridges (flip, Stirling phase, growth) exhibit the natural numbers as winding counts depending on the basepoint of the lift — the ambiguity of the representation; the symmetry cancellation above acts on the multi-valued fibers; and the phase superposition of the natural numbers is an un-locked superposition. In all three the ambiguity is the same: the covering map $t \mapsto e^{it}$ assigns $\aleph_0$ representatives to each point, and an un-locked representation of a number is ambiguous. The author's final positioning: the framework is a *representation method*, not a new number field — every object used is known mathematics ($\mathbb{R}$, S^1, $\mathbb{N}$, the quotient, the kernel of exp); the representation precisely isolates the phase-superposition ambiguity caused by the $\mathbb{N}$-iteration count (the n-th successor is the unique layer $2\pi n$, machine-verified iso 31), and constrains the phases not to superpose arbitrarily (the locking selects one representative; the result is independent of the selection, `phaseSucc_respects_equiv`). The numbers themselves are unambiguous (Peano); what needs precision is the representation, and the single-phase representation provides it: the quotient of the translation symmetry, in which each class is one point of the unit circle. Machine-verified (iso 29): $\exp(i\theta_1) = \exp(i\theta_2)$ iff $\theta_1 \sim \theta_2$ — the kernel of exp is $2\pi i\mathbb{Z}$ (`exp_phase_equiv_iff`), so the classes correspond one-to-one to the projections and the ambiguity is removed (`single_phase_class_iff_projection`).

The single-phase representation is moreover *unique up to isomorphism* (author's question: does every attempt to represent a single-phase number end up isomorphic to the framework's?): by the universal property of the quotient, any attempt — a map $f : \mathbb{R} \to X$ that is constant on phase-equivalent representatives (no phase superposition), separating (different classes give different numbers), and surjective — factors *uniquely* through the quotient, and the factor map is a bijection (machine-verified: `single_phase_field_unique`, `single_phase_field_equiv`, iso 29). Hence every phase-superposition-free representation is isomorphic to the Pat single-phase representation: the representation is canonical.

The one structure the projection necessarily loses is the layer of the point 1 (author's question: is the loss falsifiable?). Since $1 = e^{i\cdot 0} = e^{i\cdot 2\pi} = \cdots$, the fiber of 1 is $\{2\pi n : n \in \mathbb{Z}\}$ (machine-verified:

`fiber_of_one`, iso 29): $\aleph_0$ representatives collapse to the single point 1. The claim "the projection preserves all structure" is *falsifiable* — it is false: the projection is not injective ($q(0) = q(2\pi)$, $0 \neq 2\pi$; machine-verified: `projection_loses_structure`, iso 29). This loss is not a defect: what is lost is exactly the layer under the translation symmetry — the winding count, $\pi_1(S^1) \cong \mathbb{Z}$ — which is precisely what the quotient cancels (iso 29); the single-phase number field accepts the loss of layers in exchange for the removal of ambiguity.

The path problem (author's question, machine-verified iso 30)

Whether the definition is complete enough to handle the path question: from the same starting point to the same ending point, different paths — do they give different numbers? The answer is two-layered (machine-verified in the isolation file `path_problem_iso.lean`, iso 30, 8 declarations, 0 errors, 0 `sorry`). The loop family $\gamma_n(t) = e^{2\pi nt\cdot i}$, $n \in \mathbb{Z}$ (the n-fold loop from the projection 1 back to 1) all start and end at the same projection (`loopPath_start`, `loopPath_end`); their lifts $g_n(t) = 2\pi nt$ end at the layers $g_n(1) = 2\pi n$ (`loopPath_lift`, `loopPath_layer`).

1. *Quotient layer (single-phase representation): path independent.* All loops give the same number, the class of 1: the projection is the same, and the translation symmetry is cancelled (`loopPath_same_class`) — well-defined by design.

2. *Lift layer (multi-phase): path dependent.* Different loops ($n \neq m$) lift to different layers, $2\pi n \neq 2\pi m$ (`loopPath_layer_distinct`) — the path information is preserved in the layers.

The definition is therefore complete in both directions (`path_problem_complete`): the single-phase number (the quotient) is path-independent, and the multi-phase layer (the lift) is path-dependent; both are expressed, and the path problem is resolved rather than avoided.

A further calibration (author): carrying the *full* path information would require a component that is not phase. Phase (θ, n) carries at most the winding class of a path — its homotopy class, $\pi_1(S^1) \cong \mathbb{Z}$ — while the full path (its shape) is a continuous function, a different component; the lift is determined by the projection path and the starting layer, so the projection path is an independent input, not phase. The single-phase representation resolves this by being path-independent by definition (the quotient): the number carries no path information at all; the multi-phase layer carries the winding class, the maximum that phase can carry; the full path, when needed, is an extra component outside the representation. No "path-carrying phase" is required.

Successor as phase superposition (author, iso 31)

The component that carries the movement is Pat itself: the successor. Its essence is phase superposition (machine-verified in `phase_succ_iso.lean`, iso 31, 4 declarations, 0 errors, 0 `sorry`): the successor is one turn, $\mathrm{succ}(\theta) = \theta + 2\pi$ — (i) *aligned*: the phase modulation preserves the projection, $\exp(\mathrm{succ}(\theta)\cdot i) = \exp(\theta \cdot i)$ (`phaseSucc_preserves_projection`) — the modulation aligns to the successor direction of the previous operation without changing the phase point; (ii) *executed*: the layer is raised by one, from $2\pi n$ to $2\pi(n + 1)$ (`phaseSucc_layer`). The natural numbers are the phase superpositions from 0: the n-th successor is $2\pi n$, the winding number n (`phaseSucc_iterate`). The successor is thereby redefined from a Peano axiom to a structural operation — phase superposition, the generator of the translation symmetry (`phaseSucc_is_translation`).

Only 0 and 1 (author's deepest conviction)

Because the operators and the numbers are ambiguous (phase superposition, paths, basepoints), how is the continued definition of numbers regulated? The answer (machine-verified, iso 31, 9 declarations): the primitives of the framework are only 0 (the basepoint, the empty operation) and 1 (the single step, one phase superposition); every number is an iterate of 1 — each step is a single step reaching a point, and "changing the point" means continuing from the current layer, never re-selecting. The well-definedness is fourfold: the successor respects phase equivalence (`phaseSucc_respects_equiv`); the iterate is unique — the n-th successor is $2\pi n$ (`phaseSucc_iterate`); the addition is regulated — m steps after n steps is $n + m$

steps (`phaseSucc_iterate_add`); and in the quotient (the single-phase representation) the successor is the identity, $[\theta+2\pi]=[\theta]$ (`phaseSucc_trivial_on_quotient`). The single-phase representation therefore has no number growth (the projection does not change); the numbers $0,1,2,\ldots$ live in the layers and are regulated by the locking; the projections of 0 and 1 coincide (both are 1).

Phase superposition and the continuum

The same phase structure generates the continuum itself. In the framework basepaper [2] the *continuum closure theorem* is machine-verified: the countable pat grid

$$\mathrm{patGrid}=\{\,2\pi j/N \mid 0<N,\ j\le N\,\}$$

is dense in $[0,2\pi]$ (R150), and therefore the continuum $[0,2\pi]$ is the closure of the grid (R151); an endogenous variant constructs the same closure from two paired basepoints and midpoint reduction (R163). The author proposed the following reading of this theorem: the continuum arises by *phase superposition* — each grid point $2\pi j/N$ carries the $\aleph_0$ phase representatives $2\pi j/N+2\pi n$, and the superposition of countably many such phase layers ("$\aleph_0$ layers of $\aleph_0$ points") closes to the continuum.

We record the cardinal arithmetic of this reading precisely (KNOWN classical; the calibration is author calibration 8). Two different counts must be separated. The *union* of all phase layers is $\aleph_0\cdot\aleph_0=\aleph_0$ (cardinal absorption [9]): the number of layers is still countable. The *superposition* — choosing one projection at each of the $\aleph_0$ positions — is the function space $\mathbb{N}\to\mathbb{N}$, with

$$\aleph_0^{\aleph_0}=2^{\aleph_0}=\mathfrak{c}=|\mathbb{R}|,$$

the continuum itself (machine-verified in the isolation file `continuum_projection_iso.lean`, iso 28 of the artifact: 15 declarations, 0 errors, 0 `sorry`). The "more than $\aleph_0$" intuition is therefore correct for the *result*: the superposition produces $2^{\aleph_0}>\aleph_0$ objects (Cantor). The choice space at each position does not matter for the result: for every $2\le\kappa\le\mathfrak{c}$ one has $\kappa^{\aleph_0}=\mathfrak{c}$ (machine-verified; e.g. binary and decimal expansions), so whether each position carries 2, 10, $\aleph_0$, or $\mathfrak{c}$ projections, the superposition closes to the same continuum.

The four phase spaces are compared systematically, entirely from the structural formula (superposition = function space, base $\kappa^{\aleph_0}$ where κ is the per-position phase space; no reference to the definition of $\aleph_1$ — the machinery is *structural*, not definitional):

Per-position phase space P	$\#P$	superposition $\lvert P\rvert^{\aleph_0}$
natural numbers $\mathbb{N}$	$\aleph_0$	$\aleph_0^{\aleph_0}=\mathfrak{c}$
real numbers $\mathbb{R}$	$\mathfrak{c}$	$\mathfrak{c}^{\aleph_0}=\mathfrak{c}$
phase full spectrum $\mathbb{N}\to\mathbb{N}$	$\aleph_0^{\aleph_0}=\mathfrak{c}$	$\mathfrak{c}^{\aleph_0}=\mathfrak{c}$
all-dimension functions $\mathbb{R}\to\mathbb{R}$	$2^{\mathfrak{c}}$	$(2^{\mathfrak{c}})^{\aleph_0}=2^{\mathfrak{c}}$

(all four machine-verified, iso 28). The third row is the author's *phase full spectrum*: the phase of one natural number is the infinite-dimensional space of infinite phases per dimension, $\aleph_0^{\aleph_0}$, and the natural numbers are its $\aleph_0$-fold superposition, giving the triple power

$$(\aleph_0^{\aleph_0})^{\aleph_0}=\aleph_0^{\aleph_0\cdot\aleph_0}=\aleph_0^{\aleph_0}=\mathfrak{c}$$

(machine-verified: `triple_power_absorption`, iso 28): power associativity and the absorption $\aleph_0\cdot\aleph_0=\aleph_0$ collapse the three layers back into one — infinite nesting (finite or countable layers) absorbs to the continuum.

The *association* of the tower is decisive (author's question about paths between natural numbers). The parallel superposition above is the *shortest-path* reading: one phase choice per position, positions side by side. But every path from 0 to a natural number n may pass through any other natural numbers, so the phases of different natural numbers superpose *along paths*; the finite–infinite bridge allows infinite paths, the path space is $\mathbb{N}^{\mathbb{N}}$, and the nested superposition is the power tower

$$\aleph_0^{(\aleph_0^{\aleph_0})}=\aleph_0^{\mathfrak{c}}=2^{\mathfrak{c}},$$

exceeding the continuum (machine-verified: `path_superposition_tower`, iso 28). The contrast is exactly the association: $(\aleph_0^{\aleph_0})^{\aleph_0}=\mathfrak{c}$ (parallel, absorbs) vs. $\aleph_0^{(\aleph_0^{\aleph_0})}=2^{\mathfrak{c}}$ (nested, tower), machine-verified `nesting_exceeds_parallel`.

A calibration (author): the basepoint — the reference of the phase locking — may move freely, but the movement process is itself a path (a sequence of basepoints), already contained in the path space $\mathbb{N}^{\mathbb{N}}$; the nested superposition $\aleph_0^{(\aleph_0^{\aleph_0})} = 2^{\mathfrak{c}}$ therefore already includes basepoint movement, and the basepoint adds no new layer (machine-verified: `basepoint_movement_no_new_layer`, iso 28): the result remains $2^{\mathfrak{c}}$, not $2^{2^{\mathfrak{c}}}$. Only a genuinely new degree of freedom independent of paths would raise the tower (the arithmetic fact $\aleph_0^{2^{\mathfrak{c}}} = 2^{2^{\mathfrak{c}}}$ is recorded in iso 28 but is not triggered by basepoint movement).

A precise statement of what is exceeded, and what is not (author's question whether the continuum is "falsified"): the nested superposition of the polyphase natural numbers, $\aleph_0^{(\aleph_0^{\aleph_0})} = 2^{\mathfrak{c}}$, is larger than the continuum (Cantor; machine-verified) — what is falsified is the *ceiling* claim that every superposition of natural-number phases stays $\leq \mathfrak{c}$ (true for parallel superpositions, false for the nested one). The continuum itself is not falsified: $\mathfrak{c} = |\mathbb{R}| = 2^{\aleph_0}$ is a theorem (machine-verified), and the continuum hypothesis $\mathfrak{c} = \aleph_1$ is independent of ZFC, so neither is disprovable. The nested superposition equals $|\mathbb{R} \to \mathbb{R}|$, the function space above the continuum. The comparison shows: any phase space of size $\leq \mathfrak{c}$ — natural, real, or the full spectrum — superposes to the same continuum $\mathfrak{c}$ (the ceiling $\kappa^{\aleph_0} = \mathfrak{c}$ for $2 \leq \kappa \leq \mathfrak{c}$); the only superposition exceeding the continuum is the all-dimension one, where the phase space itself is $2^{\mathfrak{c}} > \mathfrak{c}$ (Cantor) — the excess comes from the per-position space, not from the number of positions or the superposition operation.

The superposition result is the second layer of the power hierarchy, $\beth_1 = \mathfrak{c} = \aleph_0^{\aleph_0}$ (machine-verified, iso 28). The aleph hierarchy $\aleph_0 < \aleph_1 < \aleph_2 < \cdots$ (ordinal-successor construction) and the beth hierarchy $\beth_0 = \aleph_0$, $\beth_1 = \mathfrak{c}$, $\beth_2 = 2^{\mathfrak{c}}$ (power construction) are *different constructions*: $\aleph_1$ is not reached by powers — it is the successor of $\aleph_0$ — and the superposition lands at $\beth_1$. Where $\beth_1$ sits in the aleph hierarchy, $\beth_1 = \aleph_1$ (the continuum hypothesis) or higher, is independent of ZFC — a question of independence, not of definition.

Two further counts calibrate the author's questions. First, the superposition result $\aleph_0 < \mathfrak{c}$ is definite (Cantor), and $\aleph_1 \leq \mathfrak{c}$ always holds; there is no cardinal between $\aleph_0$ and $\aleph_1$, but this is a constructive fact about $\aleph_1$ (it is the least uncountable cardinal, a successor by construction) — recorded here only as a calibration, not as a conclusion of the superposition. If each position carries $\aleph_1$ projections (the positions are still the $\aleph_0$ natural numbers), the result is still $\aleph_1^{\aleph_0} = \mathfrak{c}$: $\aleph_1 \leq \mathfrak{c}$ always holds ($\aleph_1$ is the least uncountable cardinal), and the ceiling $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ absorbs it (machine-verified). Only if the *positions* themselves number $\aleph_1$ does one leave the continuum construction: $\aleph_1^{\aleph_1} = 2^{\aleph_1} \geq \aleph_2$ (Cantor; machine-verified), where the equality $2^{\aleph_1} = \aleph_2$ is the generalized continuum hypothesis, independent of ZFC (GCH/Easton; KNOWN). Finally, the imaginary-unit powers

$$I^{c \cdot I} = e^{-c\pi/2} \qquad (c \in \mathbb{R}),$$

the case $c = 1/2$ being $I^{I/2} = e^{-\pi/4}$, real — encode a continuum-sized parameter space into the phase structure of each position (machine-verified: the parameter map is injective); even so the ceiling $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ is unchanged.

The continuum hypothesis itself remains independent of ZFC (Gödel [6], Cohen [7]); the phase-superposition reading is a *constructive generation* of the continuum (density + closure and the sequence-space count), not an intermediate-cardinality statement, and the present paper makes no claim about the continuum hypothesis.

12 Verification record

All source files, the hash ledger (`iso_hashes.md`), and the verification log are archived at the Zenodo records 10.5281/zenodo.22038154 (version 1), 10.5281/zenodo.22039506 (version 2, containing Layer 13 Blocks A–B), 10.5281/zenodo.22044557 (version 3), and 10.5281/zenodo.22044634 (version 4, the complete submission package including the isolation files iso 28–31); the same package is available at the GitHub repository listed in the cover letter. The journal's submission system accepts the manuscript PDF only; the complete Lean verification package is retrievable from these records. Environment: Lean 4.32.2 [10], mathlib revision 905b95818e [3]. The development comprises 231 declarations in `proof.lean` plus the isolation files `zeta_*_iso.lean`, `continuum_projection_iso.lean` (iso 28: the continuum by phase superposition, 35 declarations), and `symmetry_cancel_phase_iso.lean` (iso 29: what symmetry cancels and the single-phase number field, 18 declarations), and `path_problem_iso.lean` (iso 30: the path problem, 8 declarations), and `phase_succ_iso.lean` (iso 31: the successor as phase superposition and only-0-and-1, 9 declarations), and `zeta_odd_even_continuity_iso.lean` (iso 41: the odd/even pairing continuity of this revision, 4 declarations); each isolation file compiles independently with 0 errors and 0 `sorry`. The library theorems used by the Euler–Maclaurin continuation blocks of version 1 (all confirmed present in mathlib):

`HasDerivAt.cpow_const`, `ContinuousAt.cpow`, `differentiableOn_tsum_of_summable_norm`, `TendstoLocallyUniformlyOn.differentiableOn`, `AnalyticOnNhd.eqOn_zero_of_preconnected_of_frequently_eq_zero`, `Convex.isPreconnected`, `intervalIntegral.integral_ofReal`, `Complex.ofReal_cpow`, `ZetaAsymptotics.term_nonneg`, `ZetaAsymptotics.zeta_limit_aux1`.

13 Conclusion and outlook

Final claim. Pat is a representation tool for infinite dimensions: through phase superposition and locking it represents structures from the countable ($\aleph_0$) to the continuum ($\mathfrak{c}$) and beyond (up to $2^{\mathfrak{c}}$, including infinite-dimensional function spaces); it eliminates the ambiguity of numbers — the phase-superposition ambiguity of the representatives, removed by the locking and the quotient (iso 29) — and restores the structure lost by the projection — the layers (the winding counts) carried in the lift (iso 30); both supported by the empirical studies of the framework [2]. The claim is machine-verified in known mathematics (iso 28–31: 70 declarations, 0 errors, 0 `sorry`; every object — $\mathbb{R}$, S^1, $\mathbb{N}$, the quotient, the kernel of exp — is standard). Honest boundaries are recorded throughout: the tool is one of several possible representations (no uniqueness or novelty of the objects is claimed); the numbers themselves are unambiguous (Peano); and the continuum hypothesis remains independent of ZFC.

Along the path of known mathematics only, the Riemann direction is machine-verified through twelve layers: from the complex-axis representation and the circle geometry, through the false-real-axis framework and the critical-line phase layer, the covering-space phase locking, the predictor residuals, and the argument-principle rectangle closure with its boundary estimates. The negativity $\zeta(x) < 0$ on $(0, 1)$ is closed by the odd/even pairing of this revision: the continuity of ζ on $(0, 1)$ is machine-verified (iso 41, §10.3), and the sign follows directly from the machine-verified positivity of the pairing terms and the classical doubling identity (§10.4), completing the zero-free bottom edge of the rectangle of Layer 12.

A Complete declaration inventory (231 + 4 declarations, by layer)

1. **L1 (11)**: `add`, `neg`, `zero`, `one`, `J`, `mul`, `sub`, `lift`, `norm`, `primeCircle`, `criticalCircle`; `card_units_zmod_prime`, `fermat_little`, `fermat_little_complete`, `infinitely_many_primes`.
2. **L2 (13)**: `orthogonal_frame_crt`, `factor_observable_zero`, `distinct_primes_observable`, `prime_unit_on_other_circle`, `zeta_zero_reflection`, `zeta_zero_strip_reflection`, `zeta_zero_offline_pair`.
3. **L3 (22)**: `j`, `normSq`, `j_sq`, `normSq_mul`, `imaginary_radius_circle`, `real_radius_circle`, `orthogonal_radii`, `circles_disjoint`, `odd_prime_is_split_norm`, `critical_line_pseudonorm`, `threshold_timelike`, `threshold_spacelike`, `zero_observation_spacelike`, `recip_critical_line_hyperbola`, `imagAxis_inter_criticalCircle`, `realAxis_inter_primeCircle`, `imagAxis_inter_primeCircle`, `prime3Circle_inter_criticalCircle`, `primeCircle_inter_criticalCircle_ge5`.
4. **L4 (27)**: `j`, `normSq`, `unitCircle`, `imaginaryCircle`, `transpose`, `transpose_involutive`, `transpose_maps_one_to_i`, `transpose_origin_fixed`, `transpose_maps_unit_to_imaginary`, `transpose_maps_imaginary_to_unit`, `lattice_unique_unit`, `lattice_unique_imaginary`, `realAxis`, `imaginaryAxis`, `naturalImaginaryAxis`, `realAxis_inter_imaginaryAxis`, `realAxis_inter_unitCircle`, `realAxis_inter_imaginaryCircle`, `imaginaryAxis_inter_unitCircle`, `imaginaryAxis_inter_imaginaryCircle`, `origin_natural_coefficients`, `origin_imaginary_part`, `transpose_realAxis_eq_imaginaryAxis`, `transpose_natural_imaginary_is_natural_real`, `complex_imag_axis_coefficient`.
5. **L5 (19)**: `complex_mul_imag_leaks`, `complex_mul_cross_stays_imag`, `split_mul_imag_leaks`, `leak_sign_is_unit_square`, `unit_1_on_real_circle`, `unit_j_on_imag_circle`, `real_circle_is_direction_1`, `imag_circle_is_direction_j`, `directions_orthogonal`, `euler_plane`.
6. **L6 (9)**: `complex_zeta_convergence_region`, `cosh_lower_bound`, `cosh_unbounded`, `log_nat_unbounded`, `split_zeta_term_unbounded`, `zeta_term_norm_split`, `period_pair_reduces`, `divergence_pair_reduces`, `term_conj_symmetry`, `functional_equation_bridges`.

7. **L7 (18)**: `prime_factor_zero`, `trivial_zero_condition`, `critical_line_observation`, `i_pow_i_eq_exp_neg_pi_div_two`, `i_pow_i_ne_i`, `i_pow_i_pos_lt_one`, `axis_tick_pow`, `prime_no_power_decomposition`, `xi_symmetry_zero`, `xi_symmetry`, `xi_entire`, `zeta_series_form`, `zeta_functional_equation`, `alternating_series_converges`, `alternating_partial_sum_bounded`, `eta_pair_form`.
8. **L8 (31)**: `zeta_conj_of_one_lt_re`, `arg_two_ne_pi`, `arg_pi_ne_pi`, `conj_two_cpow`, `conj_pi_cpow`, `conj_sin_half`, `conj_gamma_one_sub`, `chi_conj`, `zeta_conj_of_re_lt_zero`, `conj_cpow_of_pos`, `conj_f_modif_zero`, `conj_mellin_f_modif`, `conj_completedRiemannZeta_zero`, `zeta_conj_of_critical_strip`, `zeta_re_conj_symm`, `zeta_im_conj_antisymm`, `zeta_re_even_on_line`, `zeta_im_odd_on_line`, `zeta_eq_zero_iff_re_im`, `riemannZeta_ne_zero_of_re_lt_zero`, `riemannZeta_ne_zero_of_re_eq_zero`, `nontrivial_zero_in_critical_strip`, `zeta_eq_odd_add_even`, `zeta_eq_zero_iff_p_split`, `p_split_mul_zero_iff`, `odd_part_extra_zero_on_imag_axis`, `zero_split_normSq`, `zero_not_on_euler_circles`, `affine_image_critical_line_is_line`, `on_line_iff_equidistant_base_one`, `critical_line_equidistant_basepoint_i`, `recip_basepoint_i_on_unit_circle`, `hardyZ_real`, `critical_line_positional`, `orbit_center_half`, `orbit_degenerate_iff_on_line`, `recip_on_critical_circle_iff`, `zero_orbit_four`.
9. **L9 (36)**: `chi_mul_chi_one_sub`, `conj_cpow_inv_half_of_unit`, `hardyZ_real`, `chi_unit_on_line`, `chi_ne_zero_on_line`, `chi_mem_slitPlane_on_line`, `zero_orbit_distinct_of_offline`, `zero_orbit_degenerate_on_line`, `zero_orbit_counting`, `zero_left_right_bijection`, `cpow_two_on_line_explicit`, `cpow_pi_on_line_explicit`, `sin_pi_quarter_add_mul_I`, `sin_pi_half_add_mul_I`, `gamma_abs_sq_on_line`, `term_on_line_explicit`, `cpow_two_half_minus_im`, `gamma_conj_arg_angle`, `gamma_doubling_arg_angle`, `gamma_reflection_arg_angle`, `gamma_quarter_phase_combine`, `zeta_neg_nat_ne_zero_of_odd`, `rh_iff_all_zeros_on_line_in_strips`, `offline_zero_gives_half_strip`, `zeta_two_arg_eq_arg_chi`, `chi_translation_two`, `chi_arg_translation_two`, `chi_translation_four`, `chi_arg_translation_four`, `zeta_eq_chi_mul_conj_on_line`, `zeta_unit_sq_eq_chi`.
10. **L10 (28)**: `zetaZeroSet`, `isClosed_zetaZeroSet`, `isDiscrete_zetaZeroSet`, `zeta_zeros_finite_in_strip`, `zetaUnit`, `continuousOn_zetaUnit`, `chi`, `continuousOn_chi_line`, `chi_ne_zero_on_line`, `chiPath`, `theta0`, `exp_theta0`, `thetaLift`, `thetaLift_lifts`, `thetaLift_zero`, `uPath`, `uLift0`, `exp_uLift0`, `uLift`, `uLift_lifts`, `uLift_zero`, `expKernel`, `expKernel_mem_iff`, `expKernel_dist_ge_two_pi`, `phase_align_two_zeta_lift_eq_chi_lift`, `zeta_unit_sq_eq_chi_short`, `zeta_lift_half_chi_lift_pi_int`.
11. **L11 (5)**: `polyError`, `poly_error_ratio`, `error_seq_predicts`, `poly_error_monotone`, `error_precision_bound`.
12. **L12 (12)**: `flip_chi_circle_bridge`, `net_flip_from_chi_lift`, `flip_count_from_theta_lifts`, `completedZeta_zero_log_reflection`, `zeta_zero_iff_completedZeta_zero_one_over_sum`, `zeta_line_two_bounded`, `Gamma_one_add_it_norm_le_one`, `zeta_ne_zero_on_bottom_edge`, `zeta_im_zero_on_real`, `zeta_neg_one_re_neg`, `bottom_edge_arg_jump`, `zeta_partial_sum_lt_integral_bound`.
13. **L13 (9, isolation file)**: `zetaEmR_N`, `zetaEmG_N`, `zetaEmG_N_eq_smul`, `zetaEmR_N_differentiableOn`, `zetaEmG_N_differentiableOn`, `zetaTermC`, `zetaTermC_fund_integral`, `zetaTermC_closed_form`, `zeta_cpow_ofReal_continuousAt` (plus `zeta_uIcc_ge`, `cpow_ofReal_hasDerivAt`, `cpow_ofReal_neg_hasDerivAt`).

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[2] Y. Wang, *Empirical Foundations of Pat: Transformer OOD Observations, Structural Laws, and Machine-Checked Continuum Theorems*, 2026, submitted for publication; framework basepaper of the 0pat release, containing the machine-verified continuum closure theorems R150–R151 (the continuum $[0, 2\pi]$ is the closure of the countable pat grid). Framework release v1: DOI 10.5281/zenodo.22038154.

[3] The mathlib community, *mathlib4*, Lean repository, `https://github.com/leanprover-community/mathlib4`, revision 905b95818e.

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[6] K. Gödel, *The Consistency of the Axiom of Choice and of the Generalized Continuum Hypothesis*, Annals of Mathematics Studies **3**, Princeton University Press, 1940.

[7] P. J. Cohen, The independence of the continuum hypothesis, *Proc. Nat. Acad. Sci. U.S.A.* **50** (1963), 1143–1148; **51** (1964), 105–110.

[8] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002 (covering spaces and $\pi_1(S^1) \cong \mathbb{Z}$, §1.3).

[9] P. R. Halmos, *Naive Set Theory*, Van Nostrand, 1960 (cardinal arithmetic; absorption $\aleph_0 \cdot \aleph_0 = \aleph_0$).

[10] L. de Moura, S. Kong, J. Avigad, F. van Doorn, and J. von Raumer, The Lean theorem prover (system description), in *CADE-25*, LNCS **9195**, Springer, 2015, pp. 378–388.

Author's proposals, calibrations, and AI contribution statement

In compliance with the Annals of Mathematics policy on AI and LLM tools, we record (i) the ideas and directions proposed by the author, each with its location in this paper; (ii) the author's calibrations of the development; (iii) the ideas contributed by AI tools; and (iv) the ideas contributed by AI that the author rejected or redirected. The author (a human) takes full responsibility for the content, its correctness, and the integrity and accuracy of its citations; AI agents are not authors.

Ideas and directions proposed by the author

1. *Critical-line circle and basepoint choice.* The author directed the study of how the critical-line circle arises in the Lean claims; the choice of basepoints for translation and inversion so that they cancel; the search for a single basepoint solving both critical-line circularization and translation–inversion cancellability; and the question whether intersection on the critical-line circle suffices, with the basepoint modified so that the odd and even parts converge to circles or straight lines meeting only on the critical-line circle (§3, §5, §4.2).

2. *Four-line splitting by $n/\log n$ and phase superposition.* The author proposed that the two odd/even lines be split into four lines by the scale $n/\log n$ to find four-line intersections, viewing the odd/even lines as phase superposition states and decomposing, from the pat0 viewpoint, how many phases are superposed in the two lines (§4.2, §6).

3. *Euler-circle compactification and the normalization question.* The author proposed that infinitely diverging frequencies be finitized by observation in the two orthogonal Euler circles (compactification into circles), and posed the equivalence of normalization with compactification into a circle (§3, §11).

4. *Symmetric continuation as difference, with alternating symmetry points.* The author proposed the symmetric continuation mechanism: alternating reflection about symmetry points (essentially a difference scheme), the non-closure of a single round of differences (the construction $1 \mapsto 3$ about 2, $3 \mapsto 0$ about 1.5, $0 \mapsto 4$ about 2, $4 \mapsto -1$ about 1.5, and so on, with a one-directional variant), the exchange of symmetry points with continuation points (each symmetry point is a continuation point of another), the consequence that any finite domain with sufficiently many elements extends to infinity by its own symmetry, and the symmetric continuation of arguments (numbers replaced by angles) (§11, §7, §10).

5. *Trigonometric cumulative continuation and the integer layers.* The author proposed that the integer layers of the argument ("mod π, drop π; the count is the lifted integer") be obtained by taking trigonometric functions and continuing by cumulative values — a Fourier phase calibration — and observed the equivalence of the count with the lifted integer, and of the lifted integer with the Riemann hypothesis itself (§7, §9).

6. *Phase modulation and phase alignment as the common mechanism.* The author proposed that the various bridges are all phase modulation and phase alignment (square-wave type), that the continuous argument is essentially phase, that continuous functions are projections of infinitely many phase-discrete components, and that a circle's diameter can serve as an axis on which infinitely discrete arc projections become continuous segments (§6, §7, §11).

7. *Finite–infinite and period–divergence conversions as $e^{i\pi}$ transforms.* The author proposed that the finite-to-infinite and infinite-to-finite conversions, and the period–divergence conversions, are all $e^{i\pi}$ transforms tied to the basepoint position and to phase modulation, with fast paths; the decomposition $e^{i\pi} + 1 = 0$ (i: orthogonal symmetry direction / dimensional lift; π: divergence–period conversion; e: finite–infinite conversion); the role of the transcendence of π in constructing the infinite of the real domain; and the recasting of framework-dependent arguments as observation in the two orthogonal Euler circles, i.e. a geometric representation of the same structure (left formally undefined) (§1.2, §11).

8. *Polyphase numbers and the necessity of single-phase numbers.* The author identified that the three bridges (flip, Stirling phase, growth) exhibit natural numbers as polyphase counts — the winding numbers of the covering map $t \mapsto e^{it}$ ($\pi_1(S^1) \cong \mathbb{Z}$) — and that the precise definition of a number requires phase locking to obtain a single-phase number, which motivates pat0 and its application to the Riemann direction (§1.2, §11).

9. *Deterministic prediction of the counting remainder.* The author proposed that the growth control $\log T$ is a deterministic prediction of the counting remainder, already solved by the pat residual method and requiring only 0pat formalization, with log as e's finite-to-infinite; and that a mapping phase relation in phase modulation can replace iterative approximation (the residual has a definite lower bound and stable direction), since iterative approximation is itself finite-reaching-infinite (§8, §**??**).

10. *Sign flips as the symmetry process; reversal docking as the period–divergence bridge.* The author proposed that oscillation is the sign-flip process of the framework (symmetry itself: a sign change with the position fixed, realized as iteration of i), that reversal docking is the period–divergence bridge, together with the continuous–discrete inverse bridge, and that the expansion of the sign-change mechanism joins the existing fast paths (§**??**, §1.2, §7).

11. *Essence of pat: cancellation of phases.* The author proposed that pat defines a single-phase number, that all numbers are phase-ambiguous multi-phase superpositions, that the method must cancel the phases by cancellation operations, and posed the question whether a symmetry group can be found that, after completion, exactly cancels the phases and leaves the theorem (§1.2, §11).

12. *From known mathematics only.* The author directed that the Riemann direction be proved from known mathematics only, without invoking unverified results of the pat framework, after the framework itself had been published at DOI 10.5281/zenodo.22038154 (§1.2).

13. *The continuum by phase superposition.* The author proposed that the continuum arises by phase superposition: countably many phase layers (each grid point carrying the $\aleph_0$ representatives $\theta + 2\pi n$, "$\aleph_0$ layers of $\aleph_0$ points") close to the continuum $[0, 2\pi]$, the strict form of which is the machine-verified continuum closure theorem of the framework basepaper [2] (R150–R151); the author further identified the superposition with the sequence space $\mathbb{N} \to \mathbb{N}$, so that the count is $\aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c} = |\mathbb{R}|$ (machine iso 28), with the calibration that the union of layers is $\aleph_0 \cdot \aleph_0 = \aleph_0$ (absorption) while the superposition itself is the power $\aleph_0^{\aleph_0}$, that each position carrying $\aleph_1$ projections still gives $\aleph_1^{\aleph_0} = \mathfrak{c}$, that $\aleph_1$ positions give $2^{\aleph_1} \geq \aleph_2$ (GCH independent), and that no intermediate-cardinality claim is made (§11).

Author's calibrations of the development

1. *No wheel-reinvention: bridge and cancel.* Asymptotic and phase arguments must be built from symmetry cancellation and exponential bridges, not from hand-derived classical expansions (§1.2, §6, §10.2).

2. *Use known curves* (§9.2).

3. *The false real axis is proved, not KNOWN* (§5, §**??**).

4. *Epistemic calibration: Lean no sorry* (§1.4).
5. *Search the library first* (mathlib `ZetaAsymptotics`; §1.5, §10.2).
6. *Symmetric continuation* of the $s > 1$ identity to $(0, 1)$ (§10.4).
7. *Polyphase numbers* (§1.2, §11).
8. *Cardinality claim calibrated.* The union of phase layers is $\aleph_0 \cdot \aleph_0 = \aleph_0$ (absorption), while the superposition (one projection per position) is the power $\aleph_0^{\aleph_0} = \mathfrak{c}$; $\kappa^{\aleph_0} = \mathfrak{c}$ for every $2 \le \kappa \le \mathfrak{c}$ (so per-position choice space is irrelevant); $\aleph_1^{\aleph_0} = \mathfrak{c}$ and $\aleph_1^{\aleph_1} = 2^{\aleph_1} \ge \aleph_2$; CH/GCH independence KNOWN ([7]); no intermediate-cardinality claim (§11).
9. *Nature of the* $(0, 1)$*-gap* (infrastructure, not mathematics; §10.4).
10. *Continuity covered by the odd/even splitting* (§10.3, §10).
11. *Writing discipline* (complete-logic organization, symbol discipline, precise intuition, legitimate presentation; throughout).

Ideas contributed by AI tools (each with its location)

1. The reduction of the $(0, 1)$ sign to the Euler–Maclaurin kernel with the identity theorem on the convex strip (technical assembly of the author's symmetric-continuation direction) (§10, §10.4).
2. The complex closed form of the Euler–Maclaurin term via integration by parts and power-law assembly (§10.2).
3. The use of mathlib's `ZetaAsymptotics` identity `zeta_limit_aux1` as the continuation seed (§1.5, §10.2).
4. The phase-lift construction via covering maps (`thetaLift`, `uLift`, `expKernel`) (§7).
5. The Weierstrass/identity-theorem toolkit choices (`differentiableOn_tsum_of_summable_norm` and `eqOn_zero_of_preconnected_of_frequently_eq_zero`) (§10.4, §12).
6. The mechanical Lean 4 code for all 231 declarations and the typesetting of this manuscript (§12, Appendix A).

Ideas contributed by AI that were rejected or redirected by the author

1. *Euler–Maclaurin continuation as the proof of continuity on* $(0, 1)$*.* The AI proposed to establish the $(0, 1)$-segment by a full Euler–Maclaurin continuation argument, including the continuity of ζ on the interval. The author rejected this: the continuity is already covered by the odd/even pairing bridge of §10.3 (both sides are locally uniform limits of discrete series — continuity by discrete superposition), and the Euler–Maclaurin kernel is needed only for the sign (§4.2, §10, §10.4).
2. *Hand-derived asymptotic expansions (Binet-type).* The AI proposed deriving Stirling-type asymptotics by classical expansions; the author rejected this as wheel-reinvention and redirected the argument to exponential bridges and known curves (§1.2, §10.2).
3. *Marking the bottom-edge real-valuedness as KNOWN.* The AI proposed citing the real-axis behavior of ζ as classical; the author corrected that the false-real-axis framework is already machine-verified and must be assembled, not cited (§5, §**??**).
4. *Difference-chain treatment of continuation.* The AI treated the symmetric continuation by difference chains; the author redirected: under pat0 a difference chain is a symmetric operation, and the continuation must be built from symmetry (oscillation about symmetry points, exchange of symmetry and continuation points) (§11).

5. *Normalization.* The AI used normalization in the geometric constructions; the author rejected it and posed instead the equivalence of normalization with compactification into a circle (§3, §11).

6. *Iterative approximation of the residual.* The AI proposed iterative approximation for the residual; the author rejected it, requiring instead a mapping phase relation with a definite lower bound and stable direction (§8).

7. *Persistence of the asymptotic route after bridging.* After the author had bridged the Stirling phase, the AI still proposed an asymptotic route; the author rejected it and required a fast path quoting known curves ("we are already bridged, why still asymptotics") (§1.2, §9.2).

8. *Paper scope limited to the four tasks.* The AI first framed the paper around the four tasks only; the author rejected this scope and required a complete review of the whole development (all 231 declarations in thirteen layers, plus the 4 declarations of the odd/even pairing of this revision) (§1, Table 1, Appendix A).

9. *Continuing the* $(0, 1)$*-segment before publication.* The AI proposed continuing the machine verification of Blocks C–G immediately; the author stopped this to secure publication first, and later closed the gap with the odd/even pairing short path instead of the Euler–Maclaurin infrastructure (§10.4).

Division of labor

The author (human) proposed the ideas and directions listed in the first subsection, designed the proof architecture and the layer decomposition, calibrated the development as listed in the second subsection, and takes full responsibility for the content of this submission, including its correctness and the integrity and accuracy of its citations. AI tools (the ZCode environment) performed the mechanical proof expansion, the writing of the Lean 4 code, and the typesetting of this manuscript; all theorems were verified by the Lean 4 compiler (0 errors, 0 `sorry`). AI agents are not authors.